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Hand-held conduit bending tool with multiple bending channels

Abstract

A hand-held conduit bending tool includes an elongated handle and a bender head disposed on the handle. The bender head comprising a body portion having at least one curved channel therein. In an embodiment, the channel is elliptical in cross-section. A handle mount is at a rear end of the body portion and rearward of the hook portion and a foot pedal is formed. In some embodiments, two channels are provided to accommodate two differently sized conduits. Markings are provided to enable use of the tool in a head-down position and in a head-up position.

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Background/Summary

FIELD OF THE DISCLOSURE
(1) The present disclosure relates to a hand-held conduit bending tool having multiple channels for manually bending electrical conduit.
BACKGROUND
(2) Hand-held conduit bending tools for bending electrical conduits are known, and usually include a one-piece body that has an elongated curved bender head at its bottom, a hook at its front end, and an upwardly projecting handle receptacle in which a shaft-like handle is removably receivable. A foot pedal is provided at the rear of the bender head, and the user can step onto the foot pedal. The bender head has a curved channel therein in which the conduit is seated and is bent during a bending operation. The hook is engaged under a length of conduit to be bent, which is held against a floor or other supporting surface, and a bending force is exerted upon the handle and the foot pedal, transmitted to the conduit by means of the hook, and curves the conduit around the bender head. Examples of such conduit bending tools are provided in U.S. Pat. No. 4,622,837 and in United States Publication Nos. 2003/0233859, 2004/0182129, and 2017/0274437.
SUMMARY
(3) A hand-held conduit bending tool in accordance with example embodiments includes an elongated handle having a free end and a connected end; and a bender head disposed on the connected end of the handle, the bender head comprising a body portion having a front end, an opposite rear end, upper and lower surfaces extending from the front end to the rear end, and opposite first and second side surfaces extending from the front end to the rear end, wherein the body portion has a curved channel formed therein extending upward from the lower surface, the channel including opposite first and second side walls Which are separated by a base wall, wherein a longitudinal axis extends from a front end of the channel to a rear end of the channel, the first side wall terminating in a first lower wall surface, the second side wall terminating in a second lower wall surface, and wherein the channel is elliptical in cross-section along all cross-sections

thereof taken transverse to the longitudinal axis, and wherein a distance taken from the base wall to each lower wall surface is less than a diameter of a conduit having a circular cross-section which is configured to be positioned within the channel.

(4) A hand-held conduit bending tool in accordance with example embodiments includes a bender head comprising a body portion having a front end, an opposite rear end, a lower surface extending from the front end to the rear end, and an opposite upper surface extending from the front end to the rear end, and a hook portion extending from the front end of the body portion, the body portion having a curved channel formed therein extending from the lower surface and from the front end to the rear end, the curved channel being configured to receive a conduit, and a handle mount provided by the body portion at the front end of the body portion and rearward of the hook portion, wherein a portion of the upper surface rearward of the handle mount forms a foot pedal; and an elongated handle having a free end and a connected end, the connected end of the handle being in the handle mount.

(5) A hand-held conduit bending tool in accordance with example embodiments includes a bender head comprising a body portion having a front end, an opposite rear end, upper and lower surfaces extending from the front end to the rear end, and opposite first and second side surfaces extending from the front end to the rear end, and a hook portion extending from the front end of the body portion, the body portion having a first curved channel formed therein extending from the lower surface and from the front end to the rear end, the first channel being proximate to the first side surface and configured to receive a first conduit, a second curved channel formed in the body portion and extending from the lower surface and from the front end to the rear end, the second channel being proximate to the second side surface and being differently sized than the first channel so as to accept a differently sized diameter conduit therein, wherein first markings are provided on the lower surface proximate to the first channel, second markings are provided on the first side surface, third markings are provided on the lower surface proximate to the second channel, and fourth markings are provided on the second side surface, wherein the first and second markings indicate bend angles for the first channel, and the third and fourth markings indicate bend angles for the second channel; and an elongated handle extending from the bender head.

(6) This Summary is provided merely for purposes of summarizing some example embodiments so as to provide a basic understanding of some aspects of the disclosure. Accordingly, it will be appreciated that the above described example embodiments are merely examples and should not be construed to narrow the scope or spirit of the disclosure in any way. Other embodiments, aspects, and advantages of various disclosed embodiments will become apparent from the following detailed description taken in conjunction with the accompanying drawings which illustrate, by way of example, the principles of the described embodiments.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) The organization and manner of the structure and operation of the disclosed embodiments, together with further objects and advantages thereof, may best be understood by reference to the following description, taken in connection with the accompanying drawings, which are not necessarily drawn to scale, wherein like reference numerals identify like elements in which:

(2) FIG. **1** depicts a side elevation view of a hand-held conduit bending tool in accordance with an embodiment of the present disclosure;

(3) FIGS. **2** and **3** depict perspective views of a bending head of the hand-held conduit bending tool;

(4) FIG. **4** depicts a top plan view of the bending head;

(5) FIG. **5** depicts a rear elevation view of the bending head;

(6) FIG. **6** depicts a rear elevation view of the bending head;
(7) FIG. **7** depicts a partial cross-sectional view of the bending head; and
(8) FIG. **8** depicts a cross-sectional view of the bending head along line **8-8** of FIG. **4**;
(9) FIG. **9** depicts a cross-sectional view of the bending head along line **9-9** of FIG. **4**;
(10) FIG. **10** depicts a cross-sectional view of the bending head along line **10-10** of FIG. **4**;
(11) FIG. **11** depicts a top perspective view of an embodiment of a design of the bending head;
(12) FIG. **12** depicts a bottom perspective view of the embodiment of the bending head of FIG. **11**;
(13) FIG. **13** depicts a front elevation view of the embodiment of the bending head of FIG. **11**;
(14) FIG. **14** depicts a rear elevation view of the embodiment of the bending head of FIG. **11**;
(15) FIG. **15** depicts a left side elevation view of the embodiment of the bending head of FIG. **11**;
(16) FIG. **16** depicts a right side elevation view of the embodiment of the bending head of FIG. **11**;
(17) FIG. **17** depicts a top plan view of the embodiment of the bending head of FIG. **11**; and
(18) FIG. **18** depicts a bottom plan view of the embodiment of the bending head of FIG. **11**.

DETAILED DESCRIPTION

(19) While the disclosure may be susceptible to embodiment in different forms, there is shown in the drawings, and herein will be described in detail, a specific embodiment with the understanding that the present disclosure is to be considered an exemplification of the principles of the disclosure, and is not intended to limit the disclosure to that as illustrated and described herein. Therefore, unless otherwise noted, features disclosed herein may be combined together to form additional combinations that were not otherwise shown for purposes of brevity. It will be further appreciated that in some embodiments, one or more elements illustrated by way of example in a drawing(s) may be eliminated and/or substituted with alternative elements within the scope of the disclosure.

(20) A hand-held conduit bending tool **20** includes a bender head **22** and an elongated handle **24**. The conduit bending tool **20** is used to bend a straight portion of the conduit **26**, **28** having a circular cross-section into a bent shape when pressure is applied to the by a user. The handle **24** has a free end **32** and a connected end **34** and the bender head **22** is disposed on the connected end **34** of the handle **24**. The bender head **22** may be constructed from cast iron, aluminum, polymers with various fillers to promote strength, or other suitable material. The handle **24** may be constructed from plastic, or metal, such as tubular steel, or other suitable material. The handle **24** may have a grip **36**, such as for example a rubber molding, or knurling provided on the free end **32** to assist in the user grip on the handle **24**. For convenience in explanation only, terms such as upper, lower are used, but do not indicate that these directions are required during use of the conduit bending tool **20**.

(21) The bender head **22** includes a body portion **38** having a front end **40**, an opposite rear end **42**, upper and lower surfaces **44**, **46** extending between the front and rear ends **40**, **42**, first and second side surfaces **48**, **50** extending between the front and rear ends **40**, **42** and the upper and lower surfaces **44**, **46**, and a hook portion **52** extending from the front end **40** of the body portion **38**. The upper surface **44** may be formed of an upper curved bar **54**, and the lower surface **46** may be formed of a lower curved bar **56**, which are connected together by a plurality of struts **58** which define windows. Alternatively, the body portion **38** may be a solid part such that the windows are eliminated.

(22) The body portion **38** has first and second downwardly facing channels **60**, **62** which extend upwardly from the lower surface **46**. Each channel **60**, **62** extends from the front end **40** to the rear end **42** of the body portion **38**, and each has an open lower end. In some embodiments and as shown, the channels **60**, **62** have different profiles which allow for bending of two different sizes of conduit **26**, **28** (one conduit has a first diameter, and the second conduit has a second diameter) which reduces the overall cost for the user, since the user can purchase a single conduit bending tool **20**, instead of two separate conduit bending tools. In some embodiments, the channels **60**, **62** have the same profile which allow for bending of two conduits having the same diameter at the same time.

(23) The first channel **60** includes opposite first and second side walls **64**, **66** which are connected by a base wall **68**. A longitudinal centerline **70**, see FIG. **8**, extends from the front end of the first channel **60** to the rear end of the first channel **60** through the midpoint of the base wall **68**. The first side wall **64** terminates in a first lower wall surface **72**, and the second side wall **66** terminates in a second lower wall surface **74**. When the wall surfaces **72**, **74** are viewed in side elevation, each wall surface **72**, **74** has a front portion **76** which extends from the front end **40** and at a constant radius, and a tail portion **78** extending tangentially from the respective front portion **76** to the rear end **42**. The first channel **60** is elliptical in cross-section, see FIG. **7**, along all points thereof taken transverse to the longitudinal centerline **70**, and the elliptical shape is constant. A distance **80**, which may be equivalent to the semi-major axis of the ellipse, taken from the midpoint of the base wall **68** to each wall surface **72**, **74** is less than a diameter of the conduit **26** which is configured to be positioned within the first channel **60**. When the first channel **60** is viewed in cross-section along the longitudinal centerline **70**, see FIG. **8**, the first channel **60** mirrors the shape of the portions **76**, **78** and has a front portion **82** which extends from the front end **40** and at a constant radius, and a tail portion **84** extending tangentially from the front portion **82** to the rear end **42**.
(24) The second channel **62** includes opposite first and second side walk **86**, **88** which are connected by a base wall **90**. A longitudinal centerline **92**, see FIG. **9**, extends from the front end of the second channel **62** to the rear end of the second channel **62** through the midpoint of the base wall **90**. The longitudinal centerline **92** is parallel to the longitudinal centerline **70**, and the channels **60**, **62** are side-by-side. The first side wall **86** terminates in a first lower wall surface **94**, and the second side wall **88** terminates in a second lower wall surface **96**. When the wall surfaces **94**, **96** are viewed in side elevation, each wall surface **94**, **96** has a front portion **98** which extends from the front end **40** and at a constant radius, and a tail portion **100** extending tangentially from the respective front portion **98** to the rear end **42**. The second channel **62** is elliptical in cross-section, see FIG. **7**, along all points thereof taken transverse to the longitudinal centerline **92**, and the elliptical shape is constant. A distance **102**, which may be equivalent to the semi-major axis, taken from the midpoint of the base wall **90** to each wall surface **94**, **96** is less than a diameter of the conduit **28** which is configured to be positioned within the second channel **62**. As shown, distance **102** is greater than distance **80**. When the second channel **62** is viewed in cross-section along the longitudinal centerline **92**, see FIG. **9**, the second channel **62** minors the shape of the portions **98**, **100** and has a front portion **104** which extends from the front end **40** and at a constant radius, and a tail portion **106** extending tangentially from the front portion **104** to the rear end **42**. The minor diameter of the ellipse that forms the second channel **62** is greater than the minor diameter of the ellipse that forms the first channel **60**. In the embodiment shown, the first channel **60** is sized to receive a first size of conduit **26** therein, and the second channel **62** is sized to receive a second size of conduit **28** therein which is differently sized than the size of the first conduit **26**. For example, the first channel **60** is sized to receive a ½″ conduit, and the second channel **62** is sized to receive a ¾″ conduit. As another example, the first channel **60** is sized to receive a 1″ conduit, and the second channel **62** is sized to receive a ¾″ conduit. These examples are not intended to be limiting.
(25) End wall surfaces **72**, **94** have the same profile and align with each other in the direction transverse to the longitudinal centerlines **70**, **92**. When the bender head **22** is placed against a surface **30** in a head down position, the bender head **22** sits level, and such that when the bender head **22** is rolled against the surface **30**, the entire length of the wall surfaces **72**, **94** contact the surface **30**. End wall surfaces **74**, **96** may also align with end wall surfaces **72**, **94** or may be shorter, as shown. Webbing **108** is provided between the walls **66**, **88** and between the front end **40** and the rear end **42**. The webbing **108** strengthens the channels **60**, **62**. Alternatively, the walls may be continuous with each other.
(26) When the conduit **26** is positioned in the first channel **60**, the center of the conduit **26** is positioned such that a portion of the conduit **26** is always exposed from the first channel **60**, see FIG. **7**, and a space **110** is provided between the outer profile of the round conduit **26** and the

elliptical first channel **60**. When the conduit **26** is being bent, the elliptical shape allows the conduit **26** to nest within the first channel **60** and the side walls **64**, **66** provide points of contact with the conduit **26**, instead of the conduit **26** bearing against the base **68** of the first channel **60**. Since the conduit **26** extends partially out of the first channel **60**, the conduit **26** is always in contact with the surface **30** against which is it bearing during a head down bending process. As a result, pressure is always applied to the conduit **26** by the surface **30** during a head down bending process for the entire profile during bending. Likewise, when the conduit **28** is positioned in the second channel **62**, the center of the conduit **28** is positioned such that a portion of the conduit **28** is always exposed from the second channel **62**, and a space **112** is provided between the outer profile of the round conduit **28** and the elliptical second channel **62**. When the conduit **28** is being bent, the elliptical shape allows the conduit **28** to nest within the second channel **62** and the side walls **86**, **88** provide points of contact with the conduit **28**, instead of the conduit **28** bearing against the base **90** of the second channel **62**. Since the conduit **28** extends partially out of the second channel **62**, the conduit **28** is always in contact with the surface **30** against which is it bearing during a head down bending process. As a result, pressure is always applied to the conduit **28** by the surface **30** for the entire profile during a head down bending process. These features results in a round cross-sectional shape of the conduit **26**, **28** after bending. In addition, when the conduit **26**, **28** is being bent, the conduit **26**, **28** may move into the space **110**, **112**. In the prior art channels which provide a circular cross-section, the conduit bears against the base of the channel which can cause the conduit to flatten during bending. In prior art bending tools, the outer profile of the conduit is within the profile of the channel and does not extend outward of the channel. As a result, the conduit sometimes flattens or wrinkles.

(27) As a result of the structure, the bender head **22** combines two channels **60**, **62** with a single ground profile. Both channels **60**, **62** maintain their respective center die radius (4.18″ for ½″ EMT conduit for channel **60** and 5.17″ for ¾″ EMT conduit for channel **62**). The wall surfaces **72**, **94** provide a common ground profile for uniform pressure on the conduit during bending.

(28) The body portion **38** includes markings **114**, **116**, **118**, **120** thereon which indicate bend angles. The markings **114**, **116**, **118**, **120** may be raised or recessed relative to the body portion **38** and may be cast into the body portion **38**. First markings **114** are provided on the second lower wall surface **74** of the second side wall **66** which partially forms the lower surface **46**, second markings **116** are provided on the side surface **48** proximate to the lower wall surface **72**, third markings **118** are provided on the side surface **50** proximate to the lower wall surface **94**, and fourth markings **120** are provided on the second lower wall surface **96** which partially forms the lower surface **46**. The first and second markings **114**, **116** indicate bend angles for the first channel **60**, and the third and fourth markings **118**, **120** indicate bend angles for the second channel **62**. The markings **114**, **116**, **118**, **120** provide for accurate head-down and head-up bending, as well as for producing stub-ups, offsets, saddles and back-to-back bends. When the conduit bending tool **20** is being used in a head-down position (the conduit **26** and/or **28** bears against the floor during bending), the markings **116**, **118** are used by the user to determine the bend angle. When the conduit bending tool **20** is being used in a head-up position (the free end **32** is against the floor and the user's foot is against the grip **36**), the markings **114**, **120** are used by the user to determine the bend angle (surface **46** is described herein as the lower surface for ease in explanation only and is not a required orientation; when the conduit bending tool **20** is being used in the head-up position, then the surfaces **72**, **74**, **94**, **96** form an upper surface of the conduit bending tool **20**). In the embodiment as shown, since the bender head **22** has two different centerline bend radii, the markings **114**, **116** are positioned differently to show the appropriate bend angle, and the markings **118**, **120** are positioned differently to show the appropriate bend angle. The markings **114**, **116**, **118**, **120** provide easy visibility for the user to determine the appropriate bend angle in the head-down position or in the head-up position. In an embodiment which has channels **60**, **62** with the same profile, the markings **114**, **116**, **118**, **120** are not positioned differently.

(29) The body portion **38** includes a handle mount **122** which attaches the handle **24** to the body portion **38**. In some embodiments, the handle mount **122** is formed by an open ended receptacle extending downward from the upper surface **44** in the body portion **38**, and the handle **24** seats within the receptacle. The handle mount **122** may be formed by one of the struts. The handle **24** may be detachable from the bender head **22** or fixedly attached to the bender head **22**. While the handle **24** is described as being mounted to the body portion **38**, in some embodiments, the handle **24** is integrally formed with the bender head **22** such that the handle mount **122** is integrally formed with the body portion **38** and the handle **24**. The handle mount **122** is positioned proximate to the front end **40** of the body portion **38** and rearward of the hook portion **52**. The handle mount **122** generally aligns with the front end **40** of the body portion **38**. In an embodiment, the handle mount **122** is positioned within the first quarter of the front portions **82**, **104** that extends from the front end **40** of the body portion **38**. As shown, the handle mount **122** is vertically above the front portions **82**, **104** and partially overlaps the channels **60**, **62**. As a result of this forward positioning of the handle mount **122**, an elongated foot pedal **124** is provided between the handle mount **122** and the rear end **42** of the body portion **38**. The foot pedal **124** provides a long and wide rest for the user's foot which increases the comfort of the user. A plurality of deep serrations may be formed in the upper surface of the foot pedal **124** for improved gripping by the boot of the user during the application of force by the user onto the foot pedal **124** during the bending operation. Since the handle mount **122** is positioned proximate to the front end **40** of the body portion **38**, additional room is provided for the user's boot on the upper surface **44** of the body portion **38** rearward of the handle mount **122** versus prior art bending tools which provide the handle mount **122** at approximately the center of the body portion **38**. As a result of the positioning, the length of the body portion **38** is the same as provided in this prior art, but since the handle mount **122** has been moved forward in the present disclosure, additional room is provided for the user's boot without adding material cost which would result from adding an extension for the foot rest. Because the foot pedal **124** is enlarged, this provides for increased toe room to ensure greater control, stability and leverage during operation. The handle mount **122** defines a central axis **126**, see FIG. **10**, which extends from its open end to its closed lower end. The central axis **126** is vertical when a 30 degree bend has been achieved. The location of the central axis **126** may be located along the length L of the bender head **22** at location a when measured from the rear end **42** and in a plane perpendicular to the central axis **126** as shown in FIG. **8**. The central axis **126** may be located along the length L of the bender head at a distance of 58%-75% (α) of the length L of the bender head **22** away from the rear end **42**. More preferably, the central axis **126** may be located along the length L of the bender head at a distance of 60%-65% (α) of the length L of the bender head **22** away from the rear end **42**. When the handle **24** is mounted within the handle mount **122**, a user can comfortably grasp the grip **36** of the handle **24** during the bending process to apply an appropriate amount of leverage by pulling backward on the handle **24**, while comfortably stepping on the foot pedal **124**. The lower wall surfaces **72**, **94** rest on the surface **30** which provides better stability in the stand alone head-down and handle-up resting position when on the ground.

(30) The hook portion **52** is generally T-shaped, and extends forwardly from the center of the handle mount **122** at the front end **40** of the body portion **38**. The handle mount **122** strengthens the hook portion **52**. The hook portion **52** has a central spine **130** which extends longitudinally from the front end **40** of the body portion **38**, a first hook **132** that extends perpendicular to the central spine **130** and from a first side of the central spine **130**, and a second hook **134** that extends perpendicular to the central spine **130** and from a second, opposite side of the central spine **130**. The first hook **132** has an open topped, generally U-shaped channel **136** extending downward from an upper surface thereof, and which extends from a front end **138** of the first hook **132** to a rear end **140** of the first hook **132**, and which longitudinally aligns with the first channel **60** in the body portion **38**. The rear end **140** of the first hook **132** is spaced from the front end **40** of the body portion **38**. The channel **136** has a base wall **142**, and opposite side walls **144**, **146** extending from

the base wall **142** to an open upper end **148**. A rear portion of the base wall **142** and the lower portions of the side walls **144**, **146** form a cylindrical shape extending from the rear end thereof **140**. When the conduit **26** is initially inserted into the conduit bending tool **20**, the conduit **26** is slid through a portion of the first channel **60** in the body portion **38**, through the rear end **140** of the channel **136**, and then into or through the channel **136**. The conduit **26** is parallel to the central spine **130** when inserted into the channels **60**, **136**. The conduit **26** rests on the base wall **142** and the cylinder formed by the base wall **142** and the lower portions of the side walls **144**, **146** provide a stable starting point for the conduit **26**. Likewise, the second hook **134** has an open topped, generally U-shaped channel **150** extending downward from an upper surface thereof, and which extends from a front end **152** of the second hook **134** to a rear end **154** of the second hook **134**, and which longitudinally aligns with the second channel **62** in the body portion **38**. The rear end **154** of the second hook **134** is spaced from the front end **40** of the body portion **38**. The channel **150** has a base wall **156**, and opposite side walls **158**, **160** extending from the base wall **156** to an open upper end **162**. A rear portion of the base wall **156** and the lower portions of the side walls **158**, **160** form a cylindrical shape extending from the rear end **154**. When the conduit **28** is initially inserted into the conduit bending tool **20**, the conduit **28** is slid through a portion of the second channel **62** in the body portion **38**, through the rear end **154** of the channel **150**, and into the channel **150**. The conduit **28** is parallel to the central spine **130** when inserted into the channels **62**, **150**. The conduit **28** rests on the base wall **156** and the cylinder formed by the base wall **156** and the lower portions of the side walls **158**, **160** provide a stable starting point for the conduit **28**. The channels **136**, **150** may have grooves (not shown) in their upper surfaces to assist in gripping the conduits **26**, **28** when placed therein. Since the hooks **132**, **134** extend outward from the central spine **130**, a smaller moment is provided during bending than when both hooks extend from the same side of a single spine.

(31) The first hook **132** has a bottom surface **164** which includes a front portion **166** that extends from the front end **138** to a rear portion **168** that extends therefrom to the rear end **140**. The rear portion **168** is planar and extends tangentially from the front portions **76**. The front portion **166** extends at an angle relative to the rear portion **168**. The second hook **134** has a bottom surface **170** which includes a front portion **172** that extends from the front end **152** to a rear portion **174** that extends therefrom to the rear end **154**. The rear portion **174** is planar and extends tangentially from the front portions **98**. The front portion **172** extends at an angle relative to the rear portion **174**. The planar rear portions **168**, **174** provide a stable surface upon which the bender head **22** rests as shown in FIG. **1**.

(32) While two channels **60**, **62** and hooks **132**, **134** are shown, a single channel and hook may be provided, or more than two channels and hooks may be provided.

(33) Many modifications and other embodiments of the disclosure set forth herein will come to mind to one skilled in the art to which these disclosed embodiments pertain having the benefit of the teachings presented in the foregoing descriptions and the associated drawings. Therefore, it is to be understood that the disclosure is not to be limited to the specific embodiments disclosed herein and that modifications and other embodiments are intended to be included within the scope of the disclosure. Moreover, although the foregoing descriptions and the associated drawings describe example embodiments in the context of certain example combinations of elements and/or functions, it should be appreciated that different combinations of elements and/or functions may be provided by alternative embodiments without departing from the scope of the disclosure. In this regard, for example, different combinations of elements and/or functions than those explicitly described above are also contemplated within the scope of the disclosure. Although specific terms are employed herein, they are used in a generic and descriptive sense only and not for purposes of limitation.

(34) While particular embodiments are illustrated in and described with respect to the drawings, it is envisioned that those skilled in the art may devise various modifications without departing from

the spirit and scope of the appended claims. It will therefore be appreciated that the scope of the disclosure and the appended claims is not limited to the specific embodiments illustrated in and discussed with respect to the drawings and that modifications and other embodiments are intended to be included within the scope of the disclosure and appended drawings. Moreover, although the foregoing descriptions and the associated drawings describe example embodiments in the context of certain example combinations of elements and/or functions, it should be appreciated that different combinations of elements and/or functions may be provided by alternative embodiments without departing from the scope of the disclosure and the appended claims.

Claims

1. A hand-held conduit bending tool for receiving first and second conduits, the first and second conduits having different diameters, the hand-held conduit bending tool comprising: a bender head comprising a body portion having a front end, an opposite rear end, a lower surface extending from the front end to the rear end, an opposite upper surface extending from the front end to the rear end, opposite first and second side surfaces extending from the front end to the rear end, a first body channel extending upward from the lower surface and extending from the front end to the rear end, the first body channel defining a longitudinal axis, a second body channel extending upward from the lower surface and extending from the front end to the rear end, the second body channel defining a longitudinal axis, the longitudinal axes being parallel to each other, the first body channel is configured to receive the first conduit and the second body channel is configured to receive the second conduit, a central spine extending longitudinally from the body portion, the central spine having opposite first and second sides, a first hook extending from the first side of the central spine, the first hook having a front end, an opposite rear end, an upper surface extending from the front end to the rear end of the first hook, an opposite bottom surface extending from the front end to the rear end of the first hook, and a first hook channel extending downward from the upper surface of the first hook and extending from the front end to the rear end of the first hook, the first hook channel configured to receive the first conduit, a second hook extending from the second side of the central spine, the second hook having a front end, an opposite rear end, an upper surface extending from the front end to the rear end of the second hook, an opposite bottom surface extending from the front end to the rear end of the second hook, and a second hook channel extending downward from the upper surface of the second hook and extending from the front end to the rear end of the second hook, the second hook channel configured to receive the second conduit, the bender head having a length defined in a plane between the rear end of the body portion and the front ends of the first and second hook hooks, and a handle mount provided by the body portion proximate to the front end thereof and rearward of the central spine, the handle mount having a central axis which longitudinally aligns with the central spine and which is located along the length of the bender head at a distance of 60% to 65% of the length away from the rear end of the body portion, wherein the upper surface rearward of the handle mount forms an enlarged foot pedal which is configured to receive a foot of a user; and an elongated handle having a free end and a connected end, wherein a central axis of the elongated handle is defined between the ends thereof and aligns with the central axis of the handle mount, the connected end of the elongated handle being in the handle mount.
2. The hand-held conduit bending tool of claim 1, wherein the handle mount is an open ended receptacle into which the elongated handle can be inserted.
3. The hand-held conduit bending tool of claim 1, wherein the lower surface of the body portion is radiused, the bottom surface of the first hook has a front portion extending from the front end thereof and a planar rear portion extending from the front portion to the rear end thereof, the planar rear portion being angled relative to the front portion and extending tangentially from the radiused body portion, and the bottom surface of the second hook has a front portion extending from the

front end thereof and a planar rear portion extending from the front portion to the rear end thereof, the planar rear portion of the second hook being angled relative to the front portion of the second hook and extending tangentially from the radiused body portion, and wherein a part of the lower surface of the body portion and the planar rear portions of the first and second hooks are in contact with a surface when the bending tool is in a stand alone head-down and handle-up resting position.

4. The hand-held conduit bending tool of claim 1, further comprising first markings provided on the lower surface of the body proximate to the first body channel, second markings provided on the first side surface of the body proximate to the first body channel, third markings provided on the second side surface of the body proximate to the second body channel, and fourth markings provided on the lower surface of the body proximate to the second body channel, wherein the first and second markings indicate bend angles for the first body channel, and the third and fourth markings indicate bend angles for the second body channel.

5. The hand-held conduit bending tool of claim 1, wherein the central axis of the elongated handle is vertical when a 30 degree bend angle has been achieved.

6. The hand-held conduit bending tool of claim 3, wherein the central axis of the elongated handle is angled relative to vertical and the elongated handle is tilted towards the front ends of the first and second hooks and away from the enlarged foot pedal when a part of the lower surface of the body portion and the planar rear portions of the first and second hooks are in contact with the surface when the bending tool is in the stand alone head-down and handle-up resting position.

7. The hand-held conduit bending tool of claim 3, wherein the first body channel includes opposite first and second side walls which are separated by a base wall, the first side wall terminating in a first lower wall surface, the second side wall terminating in a second lower wall surface, and the first body channel is elliptical in cross-section along all cross-sections thereof taken transverse to the longitudinal axis, and a first distance is defined from a midpoint of the base wall to the first lower wall surface, and a second distance is defined from the midpoint of the base wall to the second lower wall surface, each of the first and second distances are less than the diameter of the first conduit such that the first conduit extends partially out of the first body channel when positioned therein, and the second body channel includes opposite first and second side walls which are separated by a base wall, the first side wall of the second body channel terminating in a first lower wall surface, the second side wall of the second body channel terminating in a second lower wall surface, and the second body channel is elliptical in cross-section along all cross-sections thereof taken transverse to the longitudinal axis of the second body channel, and a first distance is defined from a midpoint of the base wall to the first lower wall surface of the second body channel, and a second distance is defined from the midpoint of the base wall to the second lower wall surface of the second body channel, each of the first and second distances of the second body channel are less than the diameter of the second conduit such that the second conduit extends partially out of the second body channel when positioned therein.

8. The hand-held conduit bending tool of claim 1, wherein wherein a part of the lower surface of the body portion and parts of the first and second hooks are in contact with a surface when the bending tool is in a stand alone head-down and handle-up resting position, and wherein the central axis of the elongated handle is angled relative to vertical and the elongated handle is tilted towards the front ends of the first and second hooks and away from the enlarged foot pedal when the bending tool is in the stand alone head-down and handle-up resting position.

9. A hand-held conduit bending tool for receiving first and second conduits, the first and second conduits having different diameters, the hand-held conduit bending tool comprising: a bender head comprising a body portion having a front end, an opposite rear end, a lower surface extending from the front end to the rear end, an opposite upper surface extending from the front end to the rear end, opposite first and second side surfaces extending from the front end to the rear end, a first body channel extending upward from the lower surface and extending from the front end to the rear end, and a second body channel extending upward from the lower surface and extending from the front

end to the rear end, the lower surface of the body portion being radiused, each of the first and second body channels defining a longitudinal axis, the longitudinal axes being parallel to each other, the first body channel being configured to receive the first conduit and the second body channel being configured to receive the second conduit, a central spine extending longitudinally from the body portion, the central spine having first and second opposite sides, a first hook extending from the first side of the central spine, the first hook having a front end, an opposite rear end, an upper surface extending from the front end to the rear end of the first hook, an opposite bottom surface extending from the front end to the rear end of the first hook, and a first hook channel extending downward from the upper surface of the first hook and extending from the front end to the rear end of the first hook, the first hook channel configured to receive the first conduit, a second hook extending from the second side of the central spine, the second hook having a front end, an opposite rear end, an upper surface extending from the front end to the rear end of the second hook, an opposite bottom surface extending from the front end to the rear end of the second hook, and a second hook channel extending downward from the upper surface of the second hook and extending from the front end to the rear end of the second hook, the second hook channel configured to receive the second conduit, and a handle mount provided by the body portion proximate to the front end thereof and rearward of the central spine, wherein the upper surface of the body portion rearward of the handle mount forms a foot pedal which is configured to receive a foot of a user; and an elongated handle having a free end and a connected end, wherein a central axis of the elongated handle is defined between the ends thereof, the connected end of the elongated handle being in the handle mount, wherein a part of the lower surface of the body portion and parts of the bottom surfaces of the first and second hooks are in contact with a surface when the bending tool is in a stand alone head-down and handle-up resting position, and wherein the central axis of the elongated handle is angled relative to vertical and the elongated handle is tilted towards the front ends of the first and second hooks and away from the foot pedal when the bending tool is in the stand alone head-down and handle-up resting position.

10. The hand-held conduit bending tool of claim 9, wherein the handle mount is an open ended receptacle into which the elongated handle can be inserted.

11. The hand-held conduit bending tool of claim 9, further comprising first markings provided on the lower surface of the body portion proximate to the first body channel, second markings provided on the first side surface of the body portion proximate to the first body channel, third markings provided on the second side surface of the body portion proximate to the second body channel, and fourth markings provided on the lower surface of the body portion proximate to the second body channel, wherein the first and second markings indicate bend angles for the first body channel, and the third and fourth markings indicate bend angles for the second body channel.

12. The hand-held conduit bending tool of claim 9, wherein the central axis is vertical when a 30 degree bend angle has been achieved.

13. The hand-held conduit bending tool of claim 9, wherein the first body channel includes opposite first and second side walls which are separated by a base wall, the first side wall terminating in a first lower wall surface, the second side wall terminating in a second lower wall surface, and the first body channel is elliptical in cross-section along all cross-sections thereof taken transverse to the longitudinal axis, and a first distance is defined from a midpoint of the base wall to the first lower wall surface, and a second distance is defined from the midpoint of the base wall to the second lower wall surface, each of the first and second distances are less than the diameter of the first conduit such that the first conduit extends partially out of the first body channel when positioned therein, and the second body channel includes opposite first and second side walls which are separated by a base wall, the first side wall of the second body channel terminating in a first lower wall surface, the second side wall of the second body channel terminating in a second lower wall surface, and the second body channel is elliptical in cross-section along all cross-sections thereof taken transverse to the longitudinal axis of the second body

channel, and a first distance is defined from a midpoint of the base wall to the first lower wall surface of the second body channel, and a second distance is defined from the midpoint of the base wall to the second lower wall surface of the second body channel, each of the first and second distances of the second body channel are less than the diameter of the second conduit such that the second conduit extends partially out of the second body channel when positioned therein.

14. The hand-held conduit bending tool of claim 9, wherein the bottom surface of the first hook has a front portion extending from the front end thereof and a planar rear portion extending from the front portion to the rear end thereof, the planar rear portion being angled relative to the front portion and extending tangentially from the radiused body portion, and the bottom surface of the second hook has a front portion extending from the front end thereof and a planar rear portion extending from the front portion to the rear end thereof, the planar rear portion of the second hook being angled relative to the front portion of the second hook and extending tangentially from the radiused body portion, and wherein the planar rear portions are in contact with the surface when the bending tool is in the stand alone head-down and handle-up resting position.